



SIMATS ENGINEERING



Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -3

CO3_AT3

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Course name: Computer Networks

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MCQ QUESTIONS

1. c) Fast Recovery

Explanation: Fast Retransmit immediately resends the missing segment upon receiving 3 duplicate ACKs, while Fast Recovery halves the congestion window ($ssthresh = cwnd/2$) and sets $cwnd = ssthresh + 3 \times MSS$ rather than resetting back to 1 MSS (which is Tahoe behavior).

2. b) 16 MSS

Explanation: Since $CWND(8) < SSTHRESH(16)$, TCP is in the Slow Start phase. In Slow Start, the congestion window doubles every RTT ($8 \times 2 = 16$ MSS).

3. c) Window size

Explanation: The 16-bit Window Size field specifies the receiver window (rwnd), communicating buffer capacity to the sender to prevent buffer overflow.

4. b) Buffers small segments until ACK arrives

Explanation: Nagle's algorithm solves the "small-packet problem" by buffering outgoing tinygrams until all outstanding data is acknowledged by an incoming ACK or until a full MSS segment is ready.

5. b) CWND growth rate relative to RTT

Explanation: TCP throughput is bounded by $\frac{CWND}{RTT}$. On high-RTT satellite links, each RTT takes hundreds of milliseconds, making the additive/exponential window expansion very slow to utilize available bandwidth.

6. b) DNS responses are small and connection overhead is unnecessary

Explanation: Standard DNS queries/responses fit within a single packet (< 512 bytes), making the 3-way TCP handshake unnecessary overhead for simple lookups.

7. c) Implement Forward Error Correction (FEC)

Explanation: Real-time streaming cannot tolerate the latency introduced by TCP retransmissions. FEC sends redundant parity packets at the application layer to reconstruct lost packets instantly on the fly.

8. b) 1472 bytes

Explanation: Payload = MTU(1500) – IP Header(20) – UDP Header(8) = 1472 bytes.

9. b) 0-RTT or 1-RTT connection setup

Explanation: QUIC integrates the transport (connection) and cryptographic (TLS 1.3) handshakes into a single round trip (or 0-RTT for resumed connections), significantly reducing connection establishment delay.

10. b) Reusing the same TCP connection for multiple requests

Explanation: HTTP/1.1 persistent connections keep the TCP channel open across multiple HTTP request/response cycles, avoiding repeated 3-way handshake and slow-start overheads.

11. b) A response blocks subsequent responses on the same connection

Explanation: HTTP/1.1 requires the server to send responses strictly in the order requests were received; if an earlier response is slow or delayed, all queued responses behind it are blocked.

12. c) TCP layer

Explanation: While HTTP/2 streams are multiplexed within one TCP stream, a lost TCP packet stalls the entire TCP stream buffer until retransmitted, causing TCP-level Head-of-Line blocking (solved later by QUIC/HTTP/3 over UDP).

13 a) 301

Explanation: 301 Moved Permanently tells the client and search engines that the URL has permanently changed and bookmarks/caches should be updated.

14 b) Validate with the server before using a cached copy

Explanation: Cache-Control: no-cache allows the browser to store the resource locally, but mandates sending a validation request (e.g., using ETag or If-Modified-Since) to the origin server before serving it.

15. c) AAAA

Explanation: A records map to 32-bit IPv4 addresses, while AAAA (quad-A) records map domain names to 128-bit IPv6 addresses.

16. b) The response is truncated (TC bit set)

Explanation: When a DNS response exceeds the 512-byte UDP limit (or EDNS0 buffer limit), the server sets the Truncated bit (TC = 1), signaling the resolver to retry the request via TCP.

17. b) DNS cache poisoning

Explanation: Also known as DNS spoofing, attackers race the legitimate authoritative server by spamming forged replies with matching Transaction IDs and UDP port numbers to inject malicious entries into resolver caches.

18. b) Recursive resolver query load

Explanation: A short TTL causes cached records to expire rapidly, forcing recursive resolvers and authoritative servers to handle a much higher volume of incoming queries.

19. b) DNS → TCP handshake → HTTP GET

Explanation: The browser must first resolve the IP address via DNS, establish a reliable transport channel via a TCP 3-way handshake, and then transmit the application-layer HTTP GET request.

20. b) UDP with application-layer sequencing for game state, HTTP/3 for assets

Explanation: UDP prevents TCP retransmission stalls for time-sensitive game positions/inputs, while HTTP/3 (built on QUIC/UDP) downloads assets with independent streams to resist packet-loss-induced HOL blocking.